Faiza - 26K0019 Date:

Q no: 1

Input:
- number of rows
- number of seats per row
- 0 or 1.
- k = group size [rows] [cols].

Outputs:
- Initial seating plan
- Free seating plan per row and total free
- Allocated seats (row no and seat no)
- Updated seating plan or "No seats available".

Pseudocode:

```
Begin Main.
  Input rows, cols
for i=0 To r-1.
  for j=0 to c-1.
    Input seat [i] [j]
  End for
End for
Print "  0 1 2 ... "
for i=0 to r-1.
  Print "R", i, ":"
for j=0 to c-1
  if seats[i][j] == 0 then print "-"
Else "X"
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End for
End for.

3. total_free = 0, freecount = 0
for i = 0 to r-1
for j = 0 to c-1.
if seats[i][j] == 0 then freecount++
End for
Print "row", i, "free", freecount.
totalfree = totalfree + freecount.
End for
Print "Total free", totalfree

4 Input k <= 0 or k > cols then
Print "Invalid group size".
stop
End if.

5 middle = row/2
roworder = empty
inden = 0
for distance = 0 to row-1
for i = 0 to r-1.
If ABS(i - middle) == distance then
rowordder[inden] = i
inden++
End if
End for

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End for
found = false
foundrow = -1
foundstart = -1
for r = 0 to r-1 AND found == false
    currow = row_order[r]
    count = 0
for j = 0 to c-1
    If seats[currow][j] == 0 then
        count++.
    If count == k then
        foundrow = currow
        foundstart = j - k+1
        found = true
    Break
End if
        count = 0
End if , End for , End for
~~Print updated seating plan~~
If found == true then
Print "Group of", k, "seated in row", foundrow,
"seats", foundstart, "to", foundstart + k-1,
for j = foundstart to foundstart + k-1
    seats[foundrow][j] = 1
End for
Print update seating plan
Calculate and print remaining available seats
Else
Print "No row contain k consecutive available seats"-
End if , End main
```